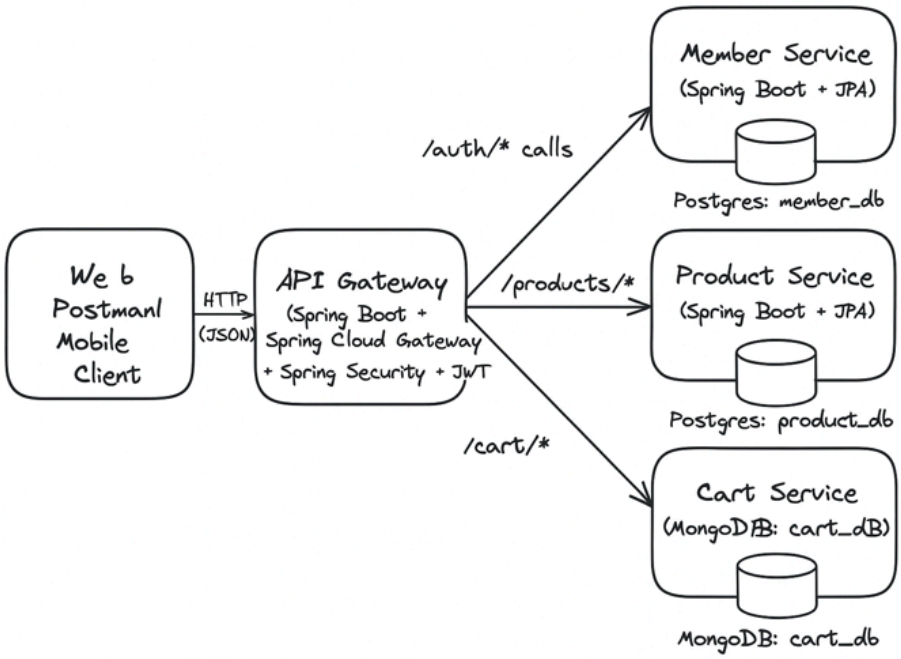
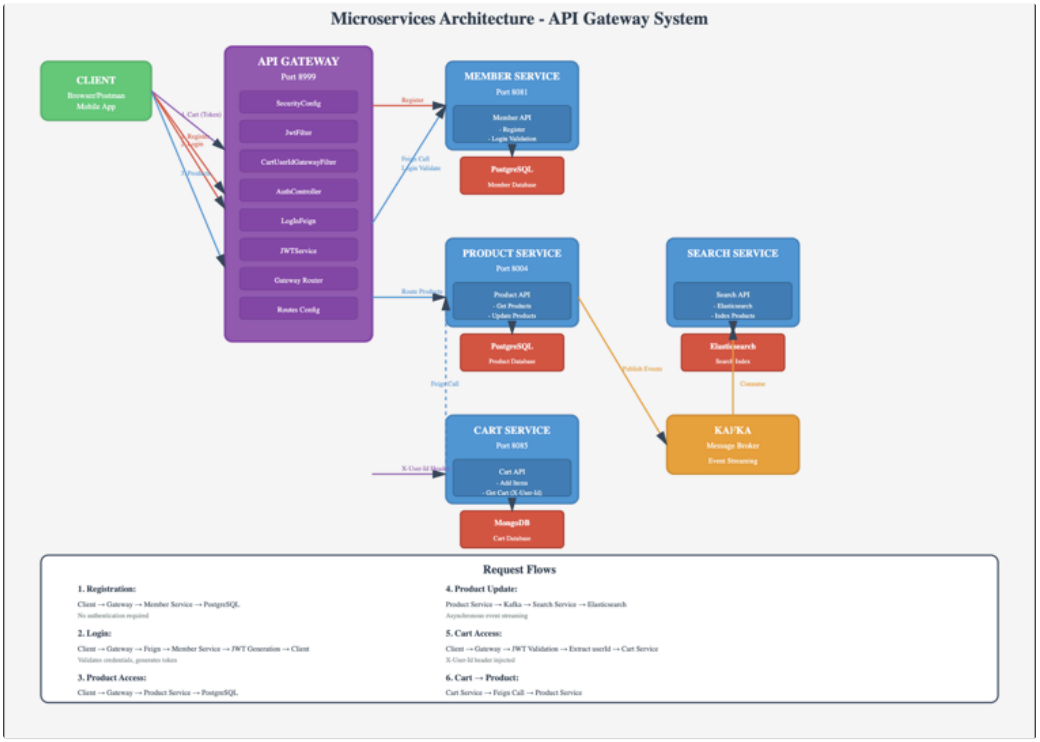
Online Market Place

Flow Diagram



Architectural Diagram



1. Member Service – PostgreSQL

1.1 Overview

- **Database**: `member_db`
- **Main table**: `members`
- Purpose: store registered customers with secure (hashed) password.

1.2 `members` table

Table name: `members`

Field	Java Type	Column Name	Constraints
id	Long	id	PRIMARY KEY, AUTO_INCREMENT
email	String	email	NOT NULL, UNIQUE, VARCHAR(255)
passwordHash	String	password_hash	NOT NULL, VARCHAR(255)
name	String	name	NOT NULL, VARCHAR(255)
phoneNumber	String	phone_number	NOT NULL, UNIQUE, VARCHAR(255)
status	String	status	NOT NULL, DEFAULT 'ACTIVE', VARCHAR(50)
createdAt	Date	created_at	NOT NULL, TIMESTAMP, AUTO
updatedAt	Date	updated_at	NOT NULL, TIMESTAMP, AUTO

2. Product Service – PostgreSQL

2.1 Overview

- **Database**: `product_db`
- **Main table**: `products`
- Purpose: store all product catalog information and support wildcard search + pagination.

2.2 `products` table

Table name: `products`

Column	Type	Constraints / Indexes	Description
`id`	`BIGINT`	`PRIMARY KEY`	Unique product identifier
`store_id`	`INT`	`NOT NULL`, `default 10001`	Default Store Id : 10001
`sku_id`	`VARCHAR(100)`	`NOT NULL`, `UNIQUE`	Unique Product code
`name`	`VARCHAR(255)`	`NOT NULL`	Product name (used in search)
`description`	`TEXT`	`NULL`	Product description
`category`	`VARCHAR(100)`	`NULL`	Category name (e.g. "Laptops")
`brand`	`VARCHAR(100)`	`NULL`	Brand name
`price`	`BIGINT`	`NOT NULL`	Price of the Product
`item_code`	`BIGINT`	`NOT NULL`	Unique item identifier

`is_active`	`BOOLEAN`	`NOT NULL`, `default TRUE`	Whether product is ACTIVE, INACTIVE
`length`	`BIGINT`	`NOT NULL`	Length of the Product
`height`	`BIGINT`	`NOT NULL`	Height of the Product
`width`	`BIGINT`	`NOT NULL`	Width of the Product
`weight`	`BIGINT`	`NOT NULL`	Weight of the Product
`dangerous_level`	`INT`	`NOT NULL`	Dangerous Level of the Product
`created_at`	`TIMESTAMP`	`NOT NULL`	Record creation timestamp
`updated_at`	`TIMESTAMP`	`NOT NULL`	Last update timestamp

Indexes:

- `idx_products_sku` on `sku_id`
- `idx_products_name` on `name`
- `idx_products_category` on `category`
- `idx_products_brand` on `brand`

---

3. Cart Service – MongoDB

3.1 Overview

- **Database**: `cart_db`
- **Collection**: `carts`
- Purpose: maintain each customer's shopping cart as a single document.

3.2 `carts` collection document structure

Collection name: `carts`

One **cart document per user**.

Document example

```
{
  "_id": "cart_12345",
  "userId": 12345,
  "items": [
    {
      "skuId": "MTA-8935-56768",
      "productId": 1001,
      "quantity": 2,
      "price": 199900,
      "addedAt": "2025-11-29T10:00:00Z"
    },
    {
      "skuId": "MTA-8935-56769",
      "productId": 1002,
      "quantity": 1,
      "price": 499900,
      "addedAt": "2025-11-29T10:05:00Z"
    }
  ],
  "totalPrice": 699800,
  "totalQuantity": 3,
  "updatedAt": "2025-11-29T10:05:00Z"
}
```

Field descriptions:

Field	Type	Description
`_id`	String	Unique cart identifier (not ObjectId)
`userId`	Long	User ID who owns this cart
`items`	Array	List of cart items
`items[].skuId`	String	Product SKU identifier
`items[].productId`	Long	Product ID
`items[].quantity`	Integer	Quantity of this item in cart
`items[].price`	Long	Price of the item (in smallest currency unit,

		e.g., cents)
`items[].addedAt`	Date	Timestamp when item was added to cart
`totalPrice`	Long	Sum of all item prices (price × quantity)
`totalQuantity`	Integer	Total number of items in cart
`updatedAt`	Date	Last update timestamp of the cart

---

4. Redis (Product)

`Cache Name`	`Purpose`	`Key Pattern`
`product`	`Individual product cache`	`product:{skuId}`
`products`	`Search results cache`	`products:search:{term}:page:{page}:size:{size}`

---

5. Search - Elasticsearch Index Schema

Field Name	Java Type	Elasticsearch Field Type	Index Name	Analyzer	Constraints / Notes
`skuId`	`String`	`_id` (Document ID)	`products`	N/A	**PRIMARY KEY** - Unique identifier, used as Elasticsearch

					document `_id`. Required, non-null.
`storeId`	`Integer`	`integer`	`products`	N/A	Store identifier. Can be null.
`name`	`String`	`text`	`products`	`standard`	Product name. **Searchable** (full-text). Normalized (lowercase, spaces removed) for space-handling. Can be null.
`description`	`String`	`text`	`products`	`standard`	Product description. **Searchable** (full-text). Normalized (lowercase, spaces removed) for space-handling.

					Can be null.
`category`	`String`	`keyword`	`products`	N/A	Product category. **Searchable** (exact match). Normalized (lowercase, spaces removed) for case-insensitive and space-handling search. Can be null.
`brand`	`String`	`keyword`	`products`	N/A	Product brand. **Searchable** (exact match). Normalized (lowercase, spaces removed) for case-insensitive and space-handling search.

					Can be null.
`price`	`Long`	`long`	`products`	N/A	Product price in smallest currency unit. Can be null.
`itemCode`	`Long`	`long`	`products`	N/A	Item code identifier. Can be null.
`isActive`	`Boolean`	`boolean`	`products`	N/A	Product active status. Can be null.
`length`	`Long`	`long`	`products`	N/A	Product length. Can be null.
`height`	`Long`	`long`	`products`	N/A	Product height. Can be null.
`width`	`Long`	`long`	`products`	N/A	Product width. Can be null.
`weight`	`Long`	`long`	`products`	N/A	Product weight. Can be null.

`dangerousLevel`	`Integer`	`integer`	`products`	N/A	Dangerous goods level. Can be null.
`createdAt`	`LocalDate`	`date`	`products`	N/A	Creation timestamp. Format: `yyyy-MM-dd`. Can be null.
`updatedAt`	`LocalDate`	`date`	`products`	N/A	Last update timestamp. Format: `yyyy-MM-dd`. Can be null.

---

1. API Gateway – Auth APIs (public)

1.1 POST /api/member/logIn

Description: Authenticates a user with email and password. The gateway validates credentials via Member Service using Feign client, and generates a JWT token upon successful authentication.

Method: `POST`

Path: `/api/member/logIn`

Authentication: Not required (public endpoint)

Request

Headers:

```
Content-Type: application/json
Accept: application/json
```

Request Body:

```
{
  "email": "user@example.com",
  "password": "userPassword123"
}
```

Request Body Schema:

Field	Type	Required	Validation	Description
`email`	String	Yes	Valid email format	User's email address
`password`	String	Yes	Not blank	User's password

Response

Success Response (200 OK):

```
{
  "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiIxMDk4MCIsImlhdCI6MTc2NDcyOTIzMiwiZXhwIjoxNzY0ODM3MjMyfQ.XcJs96in8vE4fDoKJ9uiKtB5jRKBLvyi2uaNKGbalxQ",
  "message": "Login successful"
}
```

Response Schema:

Field	Type	Description
`token`	String	JWT token (valid for 1 hour). Include in `Authorization: Bearer <token>` header for protected endpoints
`message`	String	Success message

Error Responses:

401 Unauthorized — Invalid credentials:

```
"Invalid credentials or member not found"
```

500 Internal Server Error — Server error:

```
"Login failed: <error message>"
```

---

2. Member Service APIs (Internal)

These are **not exposed directly to clients**; Gateway calls them.

2.1. POST /api/member/register

Description: Register a new member/customer with hashed password.

Endpoint: `POST /api/member/register`

Request Headers:

```
Content-Type: application/json
```

Request Body:

```
{
  "email": "user@example.com",
  "password": "StrongPassword123",
  "name": "John Doe",
  "phoneNumber": "1234567890",
  "address": "123 Main Street"
}
```

Success Response (201 Created):

```
{
  "success": true,
  "errorMessage": null,
  "errorCode": null,
  "status": 201,
  "statusText": "CREATED",
  "timestamp": "2025-12-05T10:30:00.123Z",
  "data": {
    "id": 12345,
    "email": "user@example.com",
    "name": "John Doe",
    "message": "Registered the user successfully"
  }
}
```

Error Responses:

400 Bad Request - Validation Error:

```
{
  "timestamp": "2025-12-05T10:30:00.123Z",
  "status": 400,
  "error": "Validation Error",
  "message": "Invalid email format. Email must contain a valid domain w
ith extension"
}
```

409 Conflict - Duplicate Email:

```
{
  "timestamp": "2025-12-05T10:30:00.123Z",
  "status": 409,
  "error": "Business Rule Violation",
  "message": "Email address is already registered. Please use a differe
nt email or try logging in."
}
```

409 Conflict - Duplicate Phone Number:

```
{
  "timestamp": "2025-12-05T10:30:00.123Z",
  "status": 409,
  "error": "Business Rule Violation",
  "message": "Phone number is already registered. Please use a differen
t phone number."
}
```

500 Internal Server Error:

```
{
  "timestamp": "2025-12-05T10:30:00.123Z",
  "status": 500,
  "error": "Internal Server Error",
  "message": "An unexpected error occurred. Please try again later."
}
```

2.2 POST /api/member/logIn

Description: Validate user credentials and authenticate login.

Endpoint: `POST /api/member/logIn`

Request Headers:

```
Content-Type: application/json
```

Request Body:

```
{
  "email": "user@example.com",
  "password": "StrongPassword123"
}
```

Success Response (200 OK):

```
{
  "success": true,
  "errorMessage": null,
  "errorCode": null,
  "status": 200,
  "statusText": "OK",
  "timestamp": "2025-12-05T10:30:00.123Z",
  "data": {
    "isMember": true,
    "userId": 12345
  }
}
```

Error Responses:

400 Bad Request - User Not Found:

```
{
  "timestamp": "2025-12-05T10:30:00.123Z",
  "status": 400,
  "error": "Validation Error",
  "message": "User Not Found"
}
```

400 Bad Request - Incorrect Password:

```
{
  "timestamp": "2025-12-05T10:30:00.123Z",
  "status": 400,
  "error": "Validation Error",
  "message": "Incorrect Password"
}
```

400 Bad Request - Invalid Email Format:

```
{
  "timestamp": "2025-12-05T10:30:00.123Z",
  "status": 400,
  "error": "Validation Error",
  "message": "Invalid email format. Email must contain a valid domain with extension (e.g., .com, .org)"
}
```

500 Internal Server Error:

```
{
  "timestamp": "2025-12-05T10:30:00.123Z",
  "status": 500,
  "error": "Internal Server Error",
  "message": "An unexpected error occurred. Please try again later."
}
```

---

3. Product APIs (public via Gateway)

Base: `/api/products`

3.1 GET /api/products

Description: List products with pagination

Query Parameters:

- `page` (optional, default: 0) - Page number (0-indexed)
- `size` (optional, default: 10) - Number of items per page

Example Request:

```
GET /api/products?page=0&size=20
```

Response: `200 OK`


```
{
  "success": true,
  "errorMessage": null,
  "errorCode": null,
  "status": 200,
  "statusText": "OK",
  "timestamp": "2025-12-07T10:30:02.260641Z",
  "data": {
    "content": [
      {
        "skuId": "LAP-001",
        "storeId": 10001,
        "name": "Gaming Laptop X",
        "description": "High performance gaming laptop",
        "category": "Laptops",
        "brand": "BrandX",
        "price": 799900,
        "itemCode": 123456789,
        "isActive": true,
        "length": 35,
        "height": 2,
        "width": 25,
        "weight": 2000,
        "dangerousLevel": 0,
        "createdAt": "2025-11-01T10:00:00",
        "updatedAt": "2025-11-01T10:00:00"
      }
    ],
    "page": 0,
    "size": 20,
    "totalElements": 50000,
    "totalPages": 2500
  }
}
```

3.2 GET /api/products/searchByTerm

Description: Wildcard search on product name/description + pagination. Returns empty list if no matches found.

Query Parameters:

- `searchTerm` (required) - Search term for product name/description
- `page` (optional, default: 0) - Page number (0-indexed)
- `size` (optional, default: 10) - Number of items per page

Example Request:

```
GET /api/products/searchByTerm?searchTerm=laptop&page=0&size=10
```

Response: `200 OK`


```
{
  "success": true,
  "errorMessage": null,
```

```
  "errorCode": null,
  "status": 200,
  "statusText": "OK",
  "timestamp": "2025-12-07T10:30:02.260641Z",
  "data": {
    "content": [
      {
        "skuId": "LAP-001",
        "storeId": 10001,
        "name": "Gaming Laptop X",
        "description": "High performance gaming laptop",
        "category": "Laptops",
        "brand": "BrandX",
        "price": 799900,
        "itemCode": 123456789,
        "isActive": true,
        "length": 35,
        "height": 2,
        "width": 25,
        "weight": 2000,
        "dangerousLevel": 0,
        "createdAt": "2025-11-01T10:00:00",
        "updatedAt": "2025-11-01T10:00:00"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1234,
    "totalPages": 124
  }
}
```


Error Response: `400 Bad Request` (if searchTerm is empty)


```
{
  "error": "Validation Error",
  "message": "Search term is required and cannot be empty",
  "timestamp": "2025-12-07T10:30:02.260641",
  "status": 400
}
```

3.3 GET /api/products/{skuId}

Description: Get product by SKU ID

Path Parameters:

- `skuId` (required, String) - Unique product SKU identifier

Example Request:

```
GET /api/products/LAP-001
```

Response: `200 OK`


```
{
  "success": true,
  "errorMessage": null,
  "errorCode": null,
  "status": 200,
```

```
  "statusText": "OK",
  "timestamp": "2025-12-07T10:30:02.260641Z",
  "data": {
    "skuId": "LAP-001",
    "storeId": 10001,
    "name": "Gaming Laptop X",
    "description": "High performance gaming laptop with 16GB RAM, 1TB SSD",
    "category": "Laptops",
    "brand": "BrandX",
    "price": 799900,
    "itemCode": 123456789,
    "isActive": true,
    "length": 35,
    "height": 2,
    "width": 25,
    "weight": 2000,
    "dangerousLevel": 0,
    "createdAt": "2025-11-01T10:00:00",
    "updatedAt": "2025-11-01T10:00:00"
  }
}
```


Error Response: `404 Not Found`


```
{
  "error": "Resource Not Found",
  "message": "Product not found with id: LAP-001",
  "timestamp": "2025-12-07T10:30:02.260641",
  "status": 404
}
```

3.4 POST /api/products/getSkusById

Description: Get multiple products by their SKU IDs

Request Body:


```
[
  "LAP-001",
  "LAP-002",
  "PHN-001"
]
```


Example Request:


```
POST /api/products/getSkusById
Content-Type: application/json

["LAP-001", "LAP-002", "PHN-001"]
```


Response: `200 OK`


```
{
  "success": true,
  "errorMessage": null,
  "errorCode": null,
  "status": 200,
  "statusText": "OK",
  "timestamp": "2025-12-07T10:30:02.260641Z",
```

```
  "data": [
    {
      "skuId": "LAP-001",
      "storeId": 10001,
      "name": "Gaming Laptop X",
      "description": "High performance gaming laptop",
      "category": "Laptops",
      "brand": "BrandX",
      "price": 799900,
      "itemCode": 123456789,
      "isActive": true,
      "length": 35,
      "height": 2,
      "width": 25,
      "weight": 2000,
      "dangerousLevel": 0,
      "createdAt": "2025-11-01T10:00:00",
      "updatedAt": "2025-11-01T10:00:00"
    },
    {
      "skuId": "LAP-002",
      "storeId": 10001,
      "name": "Business Laptop Y",
      "description": "Professional business laptop",
      "category": "Laptops",
      "brand": "BrandY",
      "price": 599900,
      "itemCode": 123456790,
      "isActive": true,
      "length": 33,
      "height": 2,
      "width": 23,
      "weight": 1800,
      "dangerousLevel": 0,
      "createdAt": "2025-11-01T10:00:00",
      "updatedAt": "2025-11-01T10:00:00"
    }
  ]
}
```


Error Response: `400 Bad Request` (if SKU IDs list is null or empty)


```
{
  "error": "Validation Error",
  "message": "SKU IDs list cannot be null or empty",
  "timestamp": "2025-12-07T10:30:02.260641",
  "status": 400
}
```


Error Response: `404 Not Found` (if no products found)


```
{
  "error": "Resource Not Found",
  "message": "No products found for the given SKU IDs",
  "timestamp": "2025-12-07T10:30:02.260641",
  "status": 404
}
```

3.5 GET /api/products/isPresent

Description: Check if a product exists by SKU ID

Query Parameters:

- `skuId` (required, String) - Unique product SKU identifier

Example Request:

```
GET /api/products/isPresent?skuId=LAP-001
```

Response: `200 OK`

```
{
  "success": true,
  "errorMessage": null,
  "errorCode": null,
  "status": 200,
  "statusText": "OK",
  "timestamp": "2025-12-07T10:30:02.260641Z",
  "data": true
}
```

3.6 POST /api/products/addProduct

Description: Add a new product to the catalog

Request Body:

```
{
  "skuId": "LAP-001",
  "name": "Gaming Laptop X",
  "description": "High performance gaming laptop",
  "category": "Laptops",
  "brand": "BrandX",
  "price": 799900,
  "itemCode": 123456789,
  "length": 35,
  "height": 2,
  "width": 25,
  "weight": 2000,
  "dangerousLevel": 0
}
```

Example Request:

```
POST /api/products/addProduct
Content-Type: application/json

{
  "skuId": "LAP-001",
  "name": "Gaming Laptop X",
  "description": "High performance gaming laptop",
  "category": "Laptops",
  "brand": "BrandX",
  "price": 799900,
  "itemCode": 123456789,
  "length": 35,
  "height": 2,
  "width": 25,
  "weight": 2000,
  "dangerousLevel": 0
```

```
}
```

Response: `201 Created`

```
{
  "success": true,
  "errorMessage": null,
  "errorCode": null,
  "status": 201,
  "statusText": "Created",
  "timestamp": "2025-12-07T10:30:02.260641Z",
  "data": {
    "skuId": "LAP-001",
    "storeId": 10001,
    "name": "Gaming Laptop X",
    "description": "High performance gaming laptop",
    "category": "Laptops",
    "brand": "BrandX",
    "price": 799900,
    "itemCode": 123456789,
    "isActive": true,
    "length": 35,
    "height": 2,
    "width": 25,
    "weight": 2000,
    "dangerousLevel": 0,
    "createdAt": "2025-12-07T10:30:02.260641",
    "updatedAt": "2025-12-07T10:30:02.260641"
  }
}
```

Error Response: `400 Bad Request` (validation errors)

```
{
  "error": "Validation Error",
  "message": "skuId is required",
  "timestamp": "2025-12-07T10:30:02.260641",
  "status": 400
}
```

Error Response: `409 Conflict` (duplicate SKU ID)

```
{
  "error": "Business Rule Violation",
  "message": "A product with SKU ID 'LAP-
001' already exists. Please use a different SKU ID.",
  "timestamp": "2025-12-07T10:30:02.260641",
  "status": 409
}
```

3.7 PUT /api/products/update/{skuId}

Description: Update an existing product

Path Parameters:

- `skuId` (required, String) - Unique product SKU identifier to update

Request Body:

```
{
  "skuId": "LAP-001",
  "name": "Gaming Laptop X Pro",
  "description": "Updated high performance gaming laptop",
  "category": "Laptops",
  "brand": "BrandX",
  "price": 899900,
  "itemCode": 123456789,
  "length": 35,
  "height": 2,
  "width": 25,
  "weight": 2000,
  "dangerousLevel": 0
}
```

Example Request:

```
PUT /api/products/update/LAP-001
Content-Type: application/json

{
  "skuId": "LAP-001",
  "name": "Gaming Laptop X Pro",
  "description": "Updated high performance gaming laptop",
  "category": "Laptops",
  "brand": "BrandX",
  "price": 899900,
  "itemCode": 123456789,
  "length": 35,
  "height": 2,
  "width": 25,
  "weight": 2000,
  "dangerousLevel": 0
}
```

Response: `200 OK`

```
{
  "success": true,
  "errorMessage": null,
  "errorCode": null,
  "status": 200,
  "statusText": "OK",
  "timestamp": "2025-12-07T10:30:02.260641Z",
  "data": {
    "skuId": "LAP-001",
    "storeId": 10001,
    "name": "Gaming Laptop X Pro",
    "description": "Updated high performance gaming laptop",
    "category": "Laptops",
    "brand": "BrandX",
    "price": 899900,
    "itemCode": 123456789,
    "isActive": true,
    "length": 35,
    "height": 2,
    "width": 25,
    "weight": 2000,
    "dangerousLevel": 0,
    "createdAt": "2025-11-01T10:00:00",
    "updatedAt": "2025-12-07T10:30:02.260641"
  }
}
```

Error Response: `404 Not Found` (product not found)

```
{
  "error": "Resource Not Found",
  "message": "Product not found with id: LAP-999",
  "timestamp": "2025-12-07T10:30:02.260641",
  "status": 404
}
```

Error Response: `400 Bad Request` (validation errors)

```
{
  "error": "Validation Error",
  "message": "Price is required",
  "timestamp": "2025-12-07T10:30:02.260641",
  "status": 400
}
```

3.8 DELETE /api/products/delete/{skuId}

Description: Delete a product by SKU ID

Path Parameters:

- `skuId` (required, String) - Unique product SKU identifier to delete

Example Request:

```
DELETE /api/products/delete/LAP-001
```

Response: `200 OK`

```
{
  "success": true,
  "errorMessage": null,
  "errorCode": null,
  "status": 200,
  "statusText": "OK",
  "timestamp": "2025-12-07T10:30:02.260641Z",
  "data": true
}
```

Error Response: `404 Not Found` (product not found)

```
{
  "error": "Resource Not Found",
  "message": "Product not found with id: LAP-999",
  "timestamp": "2025-12-07T10:30:02.260641",
  "status": 404
}
```

Common Error Responses

500 Internal Server Error

```
{
  "error": "Internal Server Error",
  "message": "An unexpected error occurred. Please try again later.",
  "timestamp": "2025-12-07T10:30:02.260641",
  "status": 500
}
```

409 Conflict (Data Integrity Violation)

```
{
  "error": "Data Integrity Violation",
  "message": "A product with SKU ID 'LAP-
001' already exists. Please use a different SKU ID.",
  "timestamp": "2025-12-07T10:30:02.260641",
  "status": 409
}
```

---

4. Cart APIs (protected via Gateway)

Base Path: `/api/cart`

Authentication: All endpoints require `X-User-Id` header (injected by API Gateway from JWT token)

4.1 GET /api/cart

Description: Get current user's cart (by userId from JWT via `X-User-Id` header).

Request:

- **Method:** `GET`
- **Path:** `/api/cart`
- **Headers:**
- `X-User-Id`: `{userId}` (Long, required)

Response:

- **200 OK** - Cart retrieved successfully

Response Body Example:

```
{
  "status": "SUCCESS",
  "statusText": "OK",
  "message": null,
  "data": {
    "userId": 12345,
    "items": [
      {
        "skuId": "MTA-8935-56768",
        "quantity": 2,
        "price": 799900,
        "addedAt": "2025-11-30T10:10:00Z",
        "name": "Gaming Laptop X",
```

```
        "description": "High performance gaming laptop with 16GB RAM, 1TB SSD...",
        "category": "Laptops",
        "brand": "BrandX",
        "itemCode": 1001,
        "length": 1000,
        "height": 2332,
        "width": 3232,
        "weight": 234,
        "dangerousLevel": 0
      }
    ],
    "totalQuantity": 2,
    "totalPrice": 1599800,
    "updatedAt": "2025-11-30T10:15:00Z"
  },
  "timestamp": "2025-12-07T10:15:00Z"
}
```


Empty Cart Response:


```
{
  "status": "SUCCESS",
  "statusText": "OK",
  "message": "Cart is empty",
  "data": {
    "userId": 12345,
    "items": [],
    "totalQuantity": 0,
    "totalPrice": 0,
    "updatedAt": "2025-11-30T10:15:00Z"
  },
  "timestamp": "2025-12-07T10:15:00Z"
}
```


Error Responses:

- **400 Bad Request** - Invalid userId format
- **500 Internal Server Error** - Server error

4.2 POST /api/cart/items

Description: Add an item to the current user's cart. If the item already exists, the quantity is replaced (not incremented).

Request:

- **Method:** `POST`
- **Path:** `/api/cart/items`
- **Headers:**
- `X-User-Id` : `{userId}` (Long, required)
- `Content-Type` : `application/json`
- **Request Body:**

```
{
  "skuId": "MTA-8935-56768",
  "quantity": 2
}
```

Request Body Fields:

Field	Type	Required	Description
`skuId`	String	Yes	Product SKU identifier
`quantity`	Integer	Yes	Quantity to add (must be > 0)

Response:

- **201 Created** - New item added to cart
- **200 OK** - Existing item updated in cart
- **200 OK** - Item removed (if product no longer exists in product service)
- **200 OK** - Cart is empty (if all items were removed)

Response Body (201 Created - New Item):

```
{
  "status": "SUCCESS",
  "statusText": "Created",
  "message": null,
  "data": true,
  "timestamp": "2025-12-07T10:15:00Z"
}
```

Response Body (200 OK - Updated Item):

```
{
  "status": "SUCCESS",
  "statusText": "Updated",
  "message": null,
  "data": true,
  "timestamp": "2025-12-07T10:15:00Z"
}
```

Response Body (200 OK - Cart Empty):

```
{
  "status": "SUCCESS",
  "statusText": "OK",
  "message": "Cart is empty",
  "data": {
    "userId": 12345,
    "items": [],
    "totalQuantity": 0,
    "totalPrice": 0,
```

```
    "updatedAt": "2025-11-30T10:15:00Z"
  },
  "timestamp": "2025-12-07T10:15:00Z"
}
```

Error Responses:

- **400 Bad Request** - Invalid request body (missing skuId or quantity, invalid quantity)
- **404 Not Found** - Product with given skuId does not exist (only for new items)
- **500 Internal Server Error** - Server error or product service unavailable

---

4.3 DELETE /api/cart/items/{itemId}

Description: Remove an item from the cart by item SKU ID.

Request:

- **Method:** `DELETE`
- **Path:** `/api/cart/items/{itemId}`
- **Path Parameters:**
- `itemId`: String (required) - SKU ID of the item to remove
- **Headers:**
- `X-User-Id`: `{userId}` (Long, required)

Example Request:

```
DELETE /api/cart/items/MTA-8935-56768
Headers:
  X-User-Id: 12345
```

Response:

- **200 OK** - Item removed successfully

Response Body:

```
{
  "status": "SUCCESS",
  "statusText": "OK",
  "message": "Successfully removed an item from the cart",
  "data": true,
  "timestamp": "2025-12-07T10:15:00Z"
}
```

Error Responses:

- **404 Not Found** - Cart not found for userId or item not found in cart
- **400 Bad Request** - Invalid userId format
- **500 Internal Server Error** - Server error

4.4 PATCH /api/cart/items/{itemSku}

Description: Increase or decrease the quantity of an item in the cart by a specified amount.

Request:

- **Method:** `PATCH`
- **Path:** `/api/cart/items/{itemSku}`
- **Path Parameters:**
- `itemSku`: String (required) - SKU ID of the item to update
- **Query Parameters:**
- `quantity`: Integer (required) - Amount to increase or decrease
- `operation`: String (required) - Either `"increase"` or `"decrease"` (case-insensitive)
- **Headers:**
- `X-User-Id`: `{userId}` (Long, required)

Example Request:

```
PATCH /api/cart/items/MTA-8935-56768?quantity=2&operation=increase
Headers:
  X-User-Id: 12345
```

Query Parameters:

Parameter	Type	Required	Description
`quantity`	Integer	Yes	Amount to increase/decrease (must be > 0)
`operation`	String	Yes	`"increase"` or `"decrease"` (case-insensitive)

Response:

- **200 OK** - Quantity updated successfully
- **200 OK** - Item removed (if quantity decreased to 0 or below)

Response Body (200 OK - Updated):

```
{
  "status": "SUCCESS",
```

```
  "statusText": "Updated",
  "message": "Successfully updated item quantity in the cart",
  "data": true,
  "timestamp": "2025-12-07T10:15:00Z"
}
```

Response Body (200 OK - Removed):

```
{
  "status": "SUCCESS",
  "statusText": "Removed",
  "message": "Item quantity decreased to 0 or below. Item removed from cart.",
  "data": true,
  "timestamp": "2025-12-07T10:15:00Z"
}
```

Error Responses:

- **400 Bad Request** - Invalid parameters (missing quantity/operation, invalid operation value, invalid quantity)
- **404 Not Found** - Cart not found for userId or item not found in cart
- **500 Internal Server Error** - Server error

Common Response Structure

All successful responses follow this structure:

```
{
  "status": "SUCCESS",
  "statusText": "OK" | "Created" | "Updated" | "Removed",
  "message": "string | null",
  "data": <response_data>,
  "timestamp": "ISO-8601 timestamp"
}
```

Common Error Response Structure

```
{
  "error": "Error Type",
  "message": "Detailed error message",
  "timestamp": "ISO-8601 timestamp",
  "status": <http_status_code>
}Response Wrapper Structure
```

All successful responses are wrapped in `ApiResponse<T>` format:

```
{
  "success": boolean,           // true for success, false for errors
  "errorMessage": string|null,  // Error message if failed
  "errorCode": string|null,     // Error code if failed
  "status": number,             // HTTP status code
  "statusText": string,         // HTTP status text (e.g., "OK", "CREATED")
  "timestamp": string,          // ISO 8601 timestamp
  "data": object|null           // Response payload (null if error)
```

```
}
```

---

4. Search APIs (public)

1. Full-Text Search

Endpoint: `GET /api/search`

Description: Full-text search across product name and description fields.

Response:

- **Status Code:** `200 OK`
- **Content-Type:** `application/json`


```
{
  "status": "success",
  "message": "Search completed successfully",
  "data": {
    "content": [
      {
        "skuId": "MTA-4537-43598",
        "storeId": 1,
        "name": "milk",
        "description": "freshwholemilk",
        "category": "drinks",
        "brand": "nestle",
        "price": 5000,
        "itemCode": 1234567890,
        "isActive": true,
        "length": 10,
        "height": 5,
        "width": 8,
        "weight": 250,
        "dangerousLevel": 0,
        "createdAt": "2025-12-04",
        "updatedAt": "2025-12-04"
      }
    ],
    "pageable": {
      "pageNumber": 0,
      "pageSize": 20,
      "sort": {
        "sorted": false
      }
    },
    "totalElements": 1,
    "totalPages": 1,
    "last": true,
    "first": true,
    "numberOfElements": 1,
    "size": 20,
    "number": 0,
    "empty": false
  },
  "timestamp": "2025-12-05T10:30:00"
}
```


Error Responses:

Status Code	Description	Example Response
`400 Bad Request`	Empty query parameter	`{"status": "error", "message": "Search query cannot be empty", "data": null, "timestamp": "2025-12-05T10:30:00"}`
`500 Internal Server Error`	Server error	`{"status": "error", "message": "Error performing search: [error details]", "data": null, "timestamp": "2025-12-05T10:30:00"}`